Clinic Appointment System

Objective

Build a clinic system where **patients book appointments**, **receptionists manage scheduling**, and **doctors manage consultations**, with correct **permissions, scheduling rules, and workflows**.

Roles

1. **Patient**
    - Register/login
    - View/update own profile
    - Book available appointment slots
    - View own appointments (upcoming/history)
    - Cancel (with policy) + request reschedule
    - View consultation summary after completion (read-only)
2. **Doctor**
    - View own schedule + daily queue
    - Confirm/decline appointment requests
    - Mark checked-in/completed/no-show
    - Fill consultation record (notes, diagnosis, prescription, tests)
3. **Receptionist**
    - Manage doctor schedules (availability + exceptions)
    - Confirm bookings (if clinic policy)
    - Check-in patients + manage queue order

    - Reschedule on behalf of patient
    - Cannot access medical notes
4. **Admin**
    - Manage users/roles
    - View analytics dashboard + export CSV

Required Features (MUST)

1) Doctor Availability + Slot Generation

- Weekly schedule per doctor (day, start, end)
- Exceptions: vacation/day off + one-off working day
- Slot generation based on session duration (15/30 min)

2) Booking + Conflict Rules

- Patient can book **only available slots**
- Prevent:
    - Double booking for same doctor/time
    - Overlapping appointments for the same patient
- Add **buffer time** before/after (configurable, default 5 min)

3) Appointment Lifecycle

Statuses:
`REQUESTED → CONFIRMED → CHECKED_IN → COMPLETED`
and `CANCELLED` / `NO_SHOW`

Rules:

    - Patient can cancel only if status is REQUESTED/CONFIRMED
    - Doctor or receptionist can mark NO_SHOW
    - COMPLETED requires consultation record completed

4) Rescheduling + Audit Trail

- Reschedule request by patient OR staff
- Must keep history:
    - old datetime, new datetime, changed by, reason, timestamp

5) Queue / Check-in

- Receptionist checks-in patients
- Doctor dashboard shows today queue ordered by check-in time
- Show waiting time (simple: now - check-in time)

6) EMR Lite (Consultation Record)

- For COMPLETED appointments:
    - diagnosis, notes
    - prescription items (drug, dose, duration)
    - requested tests
- Patient can view summary, not edit
- Receptionist cannot see medical notes

7) Search & Filters

- Appointment list filters:

    - status, date range, doctor, patient (staff only)

- Search by appointment id or patient name (staff only)

General rules

- DRF API for appointments + slots

Bonus Features

- For patients, implement login with social media FB, GOOGLE (OAuth, OAuth)
- Waiting list + auto-fill on cancellation (hold 30 mins)
- Payment simulation + refund policy + invoice PDF
- Telemedicine type (online link + file sharing)
- Analytics dashboard (no-show rate, peak hours, revenue)
- Unit tests > 12 with coverage report
- Implement notification system , when appointement confirmed/ cancelled etc.

Technical Constraints

- Use Django authentication + groups/permissions
- Use class-based views if reasonable
- Use database constraints to enforce unique slot booking
- Use transactions in booking/rescheduling

Submission

1. GitHub repo link

2. README: setup, sample users (patient/doctor/receptionist/admin), how to test main flows

3. Tests: run command + evidence (screenshots or CI)

General notes

- Group members up-to 5 per team
- Beautiful design is a must (You can use AI to do this task)
- You must complete the follow.
- Admin dashboard should be implemented (Don’t use the built-in one),
- Contributing to github is a MUST.

Best of luck ^^
Noha Shehab

Deadline: 2 <May 2026 (Send me the repo for your works)

All the members must commit their work to the repo.